

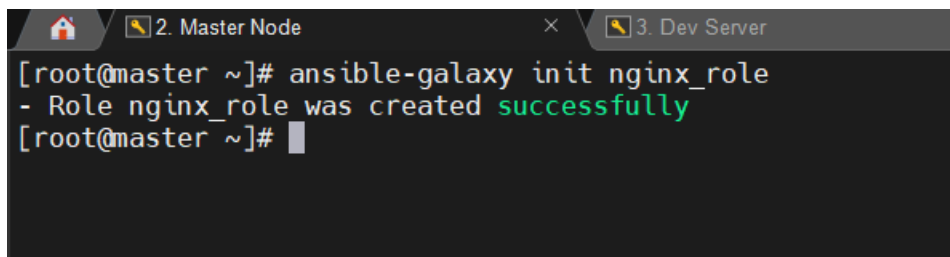
03 Assignment - Configuration

Automation using Ansible

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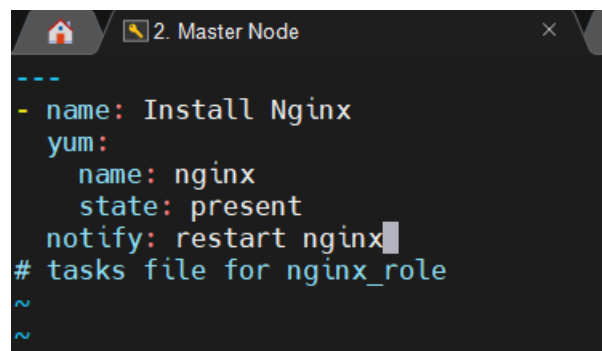
1. L3 - Create Ansible Role to define the task, handler for Nginx Service Installation and invoke the role in Ansible playbook

- Ansible Role is a **structured way** to organize playbooks.
- Instead of putting everything in one YAML file, a role has directories for tasks, handlers, defaults, files, templates
- We use ansible galaxy to create the role



```
[root@master ~]# ansible-galaxy init nginx_role
- Role nginx_role was created successfully
[root@master ~]#
```

- Define the Nginx installation task in *nginx_role/tasks/main.yml*
- The notify calls a handler if the task makes any change.



```
---
- name: Install Nginx
  yum:
    name: nginx
    state: present
    notify: restart nginx
# tasks file for nginx_role
~
~
```

- Define the handler in nginx_role/handlers/main.yml
- Handlers are triggered only if the task reports a change.

```

---
- name: restart nginx
  service:
    name: nginx
    state: restarted
# handlers file for nginx_role
~
~
~
~

```

- Create a playbook to invoke the role

```

---
- name: Install Nginx using role
  hosts: all
  become: yes
  roles:
    - nginx_role
~
~
~
~

```

```

[root@master ~]# ansible-playbook install_nginx.yml --ask-become-pass
BECOME password:
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

PLAY [Install Nginx using role] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.11.110 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
ok: [172.31.11.110]
[WARNING]: Platform linux on host 172.31.5.252 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
ok: [172.31.5.252]

TASK [nginx_role : Install Nginx] *****
ok: [172.31.11.110]
ok: [172.31.5.252]

PLAY RECAP *****
172.31.11.110      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
172.31.5.252      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
[root@master ~]#

```

- Execute the playbook and verify that Nginx is installed and running on the Worker Nodes

```
[root@Dev-Server ~]# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

- This task demonstrated the creation and use of an Ansible Role for Nginx installation. The role-based approach organized tasks and handlers efficiently, and Nginx was successfully installed and verified on the Worker Nodes.